London James Games

londonjamesgames@gmail.com

LJ Stats and Effects

July 13, 2025

Overview

LJ Stats and Effects is a Unity package developed to help speed up prototyping. In the system there are stats and effects. The 'stats' are values that each affected object has, and 'effects' are rules for how to modify the stat. It is a scriptable-object based system, but monobehaviour components are also used.

Set-Up

To set up the system, you must follow these steps.

Step 1: Create Stats

Right click in your intended directory and find 'LJ' in the create menu. You can then create an instance of a 'stat.' Feel free to modify it how you desire. Please ensure your max value is above your min value.

Step 2: Create Effects

Right click in your intended directory and find 'LJ' in the create menu. You can then create an instance of a 'StatEffect.'

Effects have a list of data elements 'data set.' You can create data elements in order to target different stats with one effect.

Stat modifiers are used to change the stat. There are some options, Add, Multiply, Set. Add adds the float value amount to the stat each time it is triggered. Multiply multiplies the stat by the float value, and set just sets it directly.

'Is Continuous' is a bool that determines whether the effect is applied only one time, or if it repeats.

The interval is the time in seconds when the effect is applied, for example, if the interval is 2, then the effect will be applied every 2 seconds, this interval element does not matter if the effect is not continuous.

Some effects may also provide immunity to other effects, simply add the effect to 'provides immunity to' in a data element and the object will not be able to acquire that effect while the effect providing immunity is active.

The 'effect type' allows you to group together effects and respond to effects of the same type in the same way.

Step 3: Create a Stat Set

Right click in your intended directory and find 'LJ' in the create menu. You can then create an instance of a 'stat set.'

Add all your intended stats and starter effects in their correct place, and this step is complete.

Step 4: Add a Local Stat Handler

In this local stat handler, assign your stat set to the 'data' section.

If you would like to add additional effects to the 'start effects' list, you may.

Step 5: Add listeners

On a child of the object with the local stat handler, add either a 'Stat Change Listener' or an 'Effect Change Listener.'

A stat change listener subscribes to the events of the Local Stat Handler, and invokes a 'response' UnityEvent. The values received from the local stat handler are normalized between 0 and 1. 0 means the value is at its minimum, while 1 means the value is at its maximum.

An ‘Effect Change Listener’ is a bit different, it subscribes to events from the local stat handler as well, however it is listening for effects. When a continuous effect is added, it will trigger the ‘effect started response’, similarly it will trigger the ‘effect ended response’ when it is removed. When the continuous effect repeats (after interval,) it will trigger the ‘effect repeated response.’ When a non-continuous effect is added, it will trigger the ‘Effect Instant Response.’

All of these responses send this thing called an ‘EffectContext’ An EffectContext is essentially just an object that keeps track of any data added to the context properties list, and the sender. This object can be accessed and its data can be used to have responses to effects that are aware of extra context.

Step 6: Create Responders and Evaluators

Similarly to the listeners, responses should be on a child object of the local stat handler. The responses to an effect should be triggered immediately in the ‘effect change listener.’

However, if you intend to trigger different actions as stats change, a ‘Progressive Responder’ is the best way to go about it. This allows you to define different stages of a stat. To use the progressive responder, drag and drop it into the response of the stat change listener. These responses will be triggered when the stat is above the floor value, and below the ceiling value.

A stat evaluator is like a filtered responder. It can be used to convert values along the stat to values on the curve, or it can be used to multiply the stat by the curve. This allows you to have different responses and send the stats value with a modifier. Please ensure that on your curve, x=0 represents the current value of the stat being its minimum, and x = 1 represents the current value of the stat being its maximum. This prevents unexpected behaviour, as the stats are normalized between 0 and 1 to other components.

Step 6: Apply Effects

You can apply effects using some of the built in components, like ‘apply stat effect on trigger enter,’ or you can alternatively create your own ways to apply stats using the static class “StatApplierUtility” and its functions.

In order to apply an effect, you can create an instance of an EffectContext in the script you apply the effect with using its constructor. Please look at Demo Scene 4 in order to see the implementation.

In order to remove an effect, you just input the EffectContext and call the ‘try remove effect’ function on the handler. Or alternatively, the remove effect by sender function and pass in the sender.

Details

Version 2.0